National Comprehensive Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

Metastatic Breast Cancer

Presented with support from

NATIONAL COMPREHENSIVE CANCER NETWORK® FOUNDATION Guiding Treatment. Changing Lives.

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NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

These NCCN Guidelines for Patients are based on the NCCN Guidelines® for Breast Cancer, Version 3.2026 — May 8, 2026.

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In memory of Judy Koh

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NCCN Foundation seeks to support the millions of patients and their families affected by a cancer diagnosis by funding and distributing NCCN Guidelines for Patients. NCCN Foundation is also committed to advancing cancer treatment by funding the nation's promising doctors at the center of innovation in cancer research. For more details and the full library of patient and caregiver resources, visit NCCN.org/patients.

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About metastatic breast cancer

Metastatic breast cancer is breast cancer that has spread to other parts of the body. The goal of treatment is to slow the spread of cancer. This guide explains what you need to know about metastatic breast cancer treatment.

What is metastatic breast cancer?

Breast cancer starts in the cells of the breast. Over time, these cells form a mass called a primary tumor. Breast cancer cells can spread through lymph or blood to other parts of the body. It's possible to develop distant metastases even when the lymph nodes near the breast are negative for cancer.

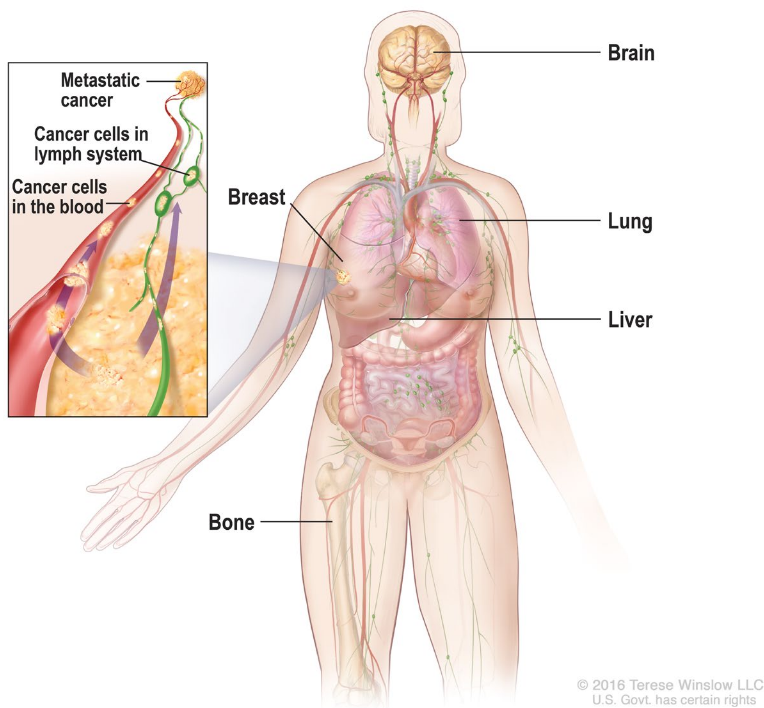
Breast cancer can metastasize almost anywhere but most commonly spreads to bone (including the spine), lung, liver, brain, or distant lymph nodes. Breast cancer that has metastasized to other parts of the body is still called breast cancer (or metastatic breast cancer) and is different from a cancer that starts in another area of the body.

How is metastatic breast cancer treated?

Metastatic breast cancer is treated with drug (systemic) therapy that travels in the bloodstream to kill cancer cells throughout the body. The goal of treatment is to slow the spread of cancer. The choice of therapy is based on hormone receptor and HER2 status of the tumor, other unique tumor features or biomarkers, your overall health, and shared decision-making between you and your care

Metastatic breast cancer

In metastatic breast cancer, cancer cells have spread through lymph or blood to other parts of the body. The most common sites are bone, lung, liver, brain, and distant lymph nodes.



team. Sometimes radiation therapy or surgery is performed if a particular site causes symptoms or other concerns.

Anyone can develop breast cancer, including people assigned male at birth. Although there are some differences between breast cancers in people assigned male and those assigned female, treatment is very similar for all genders.

How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters.

Keep reading to find the best care for you.

How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team.

2

Testing for metastatic breast cancer

Not all breast cancers are the same. Treatment planning starts with testing. It takes time for your care team to gather all the information needed to treat your cancer. This chapter presents an overview of the tests you might receive and what to expect.

Breast cancer treatment is becoming more individualized, and it's important to get the right testing. This takes time but is very helpful in making sure you get the best treatment for your type of breast cancer. A tumor biopsy and imaging studies are an important part of testing. Tests you might have are listed in **Guide 1**.

General health tests

Medical history

A medical history is a record of all health issues and treatments you've had in your life. It asks about past surgeries and current health conditions. Bring a list of old and new medicines and any over-the-counter medicines, herbals, or supplements you take.

- Some cancers are fed by hormones, such as estrogen. Therefore, it's important to let your medical team know if you're taking any hormones by mouth, through injections, or by placement of a hormone pellet. With a breast cancer diagnosis, hormones such as these will need to be discontinued.

Guide 1 Possible tests
Medical history and physical exam
Complete blood count and comprehensive metabolic panel, including liver function tests and alkaline phosphatase
Imaging tests • Chest CT. Contrast might be used • CT or MRI of abdomen with or without pelvis. Contrast will be used • Other imaging tests as needed, such as x-ray, bone scan, or PET/CT
Biopsy with tumor biomarker and testing including: • Estrogen receptor (ER) and progesterone receptor (PR) status • HER2 status • Other biomarker and mutation testing for targeted therapies
Genetic counseling and testing if at risk of hereditary breast cancer
Distress screening

Make sure you tell your care team about any symptoms you have.

Family history

Some cancers and other diseases can run in families. Your doctor will ask about the health history of family members who are blood relatives. This information is called a family history. Ask family members on both sides of your family if they had or have cancer and at what age they were diagnosed. It's important to know their specific type of cancer, where the cancer started, if it's in multiple locations, and if they had genetic testing.

Physical exam

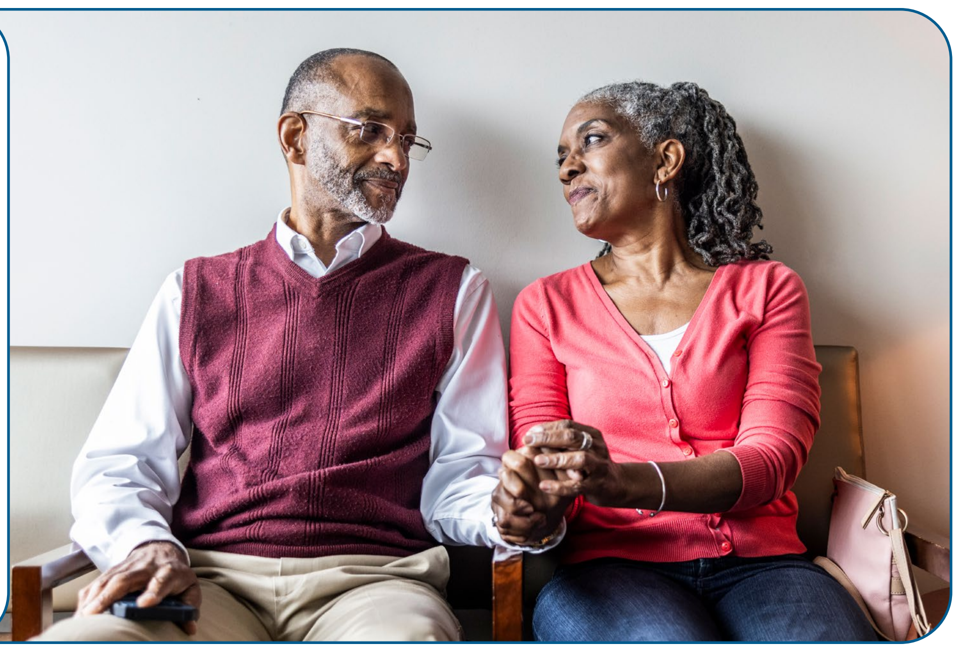
During a physical exam, your health care provider may:

- Feel and apply pressure to parts of your body to see if organs are a normal size, are soft or hard, or cause pain when touched.
- Examine your breasts to look for lumps, nipple discharge or bleeding, or skin changes.
- Feel for enlarged lymph nodes in your neck, armpit, and groin.
- Examine your spine and back

Distress screening

Dealing with a cancer diagnosis can be stressful and may cause further distress. Distress is an unpleasant experience of a mental, physical, social, or spiritual nature. It can affect how you feel, think, and act. Distress might include feelings of sadness, fear, helplessness, worry, anger, and guilt. You may also have depression, anxiety, and sleep issues. Your treatment team will screen your level of distress. This is part of your cancer care.

"It's normal to feel anxious or depressed, and there are things that can help. Make sure your support people also have access to help."



Fertility (all genders)

Treatment with systemic therapy can affect your fertility, or the ability to have children. If you think you want children in the future, ask your care team how cancer treatment might affect your fertility.

Preventing pregnancy during treatment

Preventing pregnancy during treatment is important. Cancer treatment can affect the ovaries, damage sperm, and hurt a developing baby. Therefore, becoming pregnant or having one's partner become pregnant during treatment should be avoided. If you're pregnant or breastfeeding at the time of your cancer diagnosis, treatments may need to be adjusted.

Blood tests

Blood tests check for signs of disease and how well organs are working. Some blood tests you might have are described next.

Complete blood count

A complete blood count measures the levels of red blood cells, white blood cells, and platelets in your blood. Red blood cells carry oxygen throughout your body, white blood cells fight infection, and platelets control bleeding.

Comprehensive metabolic panel

A comprehensive metabolic panel measures substances in your blood. It provides important information about how well your kidneys and liver are working, among other things

Liver function tests

Liver function tests look at the health of your liver by measuring the chemicals it makes or processes. Levels that are too high or low signal that the liver isn't working well or that cancer has spread to the liver.

Alkaline phosphatase level

Alkaline phosphatase (ALP) is an enzyme found in the blood. High levels of ALP can be a sign that cancer has spread to the bone or liver. A bone scan might be done if you have high levels of ALP.

Pregnancy test

People who can become pregnant will be given a pregnancy test before treatment begins.

Liquid biopsy or blood-based ctDNA testing for tumor mutations

Cancer cells, like all cells, shed tiny fragments of their DNA into the bloodstream. A circulating tumor DNA (ctDNA) test is a blood test that looks for cancer-specific DNA fragments. It's sometimes called a liquid biopsy because it provides information about the cancer without needing a tissue biopsy. These tests help doctors identify specific, useful targets and select medicines based on those targets.

Imaging tests

Imaging tests take pictures of the inside of your body. These tests show the primary tumor, or where the cancer started, and look for cancer in other parts of the body.

A radiologist, a medical expert, will interpret the tests and send a report to your doctor. While these reports might be available to you through your patient portal or patient access system, please wait to discuss these results with your care team. They will be able to explain what the results mean for your care.

The following imaging tests aren't in order of importance. You won't have all of these tests.

Bone x-ray

An x-ray uses low-dose radiation to take 1 picture at a time. A tumor changes the way radiation is absorbed and shows up on the x-ray. X-rays are also good at showing bone issues. Your care team may order x-rays if your bones hurt or were abnormal on a bone scan.

Contrast material

Contrast material is a substance used to improve the quality of imaging tests, such as CT and MRI scans. It's used to make the pictures clearer. Not all imaging tests require contrast, but many do.

Contrast might be taken by mouth (oral) or given through a vein (intravenously, or IV). The types of contrast vary and are different for CT and MRI.

Tell your care team if you've had allergic reactions to contrast in the past. This is important. You might be given medicines to avoid the effects of those allergies. Contrast might not be used if you have a serious allergy or if your kidneys aren't working well.

CT scan

A CT, or CAT, scan uses x-rays and computer technology to take pictures of the inside of the body. It takes many x-rays of the same body part from different angles. All the images are combined to make 1 detailed picture. IV contrast is often used.

MRI scan

An MRI scan uses radio waves and powerful magnets to take pictures of the inside of the body. MRI doesn't use radiation. Because of the strong magnets used in the MRI machine, tell the technologist if you have any metal or a pacemaker in your body. Contrast is often used.

MRI scans take longer than CT scans. A closed MRI has a capsule-like design where the magnet surrounds you. The space is small and enclosed. An open MRI has a magnetic top and bottom, which allows for an opening on each end. Closed MRIs are more common than open MRIs, so if you have claustrophobia (a fear of enclosed spaces), be sure to tell your care team.

- **A spine or brain MRI** can be used to detect breast cancer that has spread (metastasized) to the spine or brain.

Nuclear medicine bone scan

A nuclear medicine bone scan uses a radiotracer to highlight areas of bone damage or loss. A radiotracer is a substance that releases small amounts of radiation. Before the pictures are taken, the tracer is injected into your vein. It can take a few hours for the tracer to enter your bones.

A special camera will take pictures of the tracer in your bones. Areas of bone damage take up more radiotracer than healthy bone and show up as bright spots on the pictures. Bone damage can be caused by cancer, cancer treatment, previous injuries, or other health issues.

PET scan

A PET scan uses a radioactive substance called a tracer. A tracer is injected into a vein to see where cancer cells are in the body and how much sugar they're taking up. This gives an idea about how fast the cancer cells are growing. Cancer cells show up as bright spots on PET scans. However, not all tumors appear on a PET scan. Also, not all bright spots found on the PET scan are cancer. It's normal for the brain, heart, kidneys, and bladder to be bright on PET. Inflammation or infection can also show up as a bright spot.

When a PET scan is combined with CT, it's called a PET/CT scan. Types include:

- **FDG-PET/CT** uses a radiotracer called fluorodeoxyglucose (FDG). This scan is most helpful when other imaging results are unclear.
- **Sodium fluoride PET/CT** uses a radiotracer made of sodium fluoride.
- **FES-PET/CT** uses FES, which is a radioactive form of estrogen. An FES-PET/CT might be used when cancer is estrogen receptor-positive (ER+).

"My doctor reminded me that stress cannot help me, but it can hurt me! It helped me remember to control those things I could control and relax in front of those that I could not control."

Ultrasound

An ultrasound uses high-energy sound waves to form pictures of the inside of the body. This is similar to the sonogram used for pregnancy. Gel will be applied to your bare breast and a wand-like probe (transducer) will be held and moved around your bare breast and near your armpit. Ultrasound doesn't use x-rays, so it can be repeated as needed. Ultrasound is good at showing small areas of cancer near the skin. Sometimes, a breast ultrasound or MRI is used to guide a biopsy.

Biopsy

A biopsy removes a tissue sample from your body for testing. A pathologist will examine the biopsy for cancer and write a report called a pathology report. While these reports might be available to you through your patient portal or patient access system, please wait to discuss these results with your care team. They will be able to explain what the results mean for your care.

There are different types of biopsies. Some biopsies are guided using imaging, such as ultrasound or MRI. Tumors found in different areas may be biopsied. You may have tissue removed from the breast, lymph nodes, or both.

Types of biopsies include:

- **Fine-needle aspiration** and **core needle biopsy** use needles of different sizes to remove a sample of tissue or fluid.
- **Vacuum-assisted core biopsy** uses suction through a needle to remove the sample.

Since your type of breast cancer may change over time, a repeat biopsy may be needed to guide treatment.

Before a biopsy, the area is injected with numbing medicine. A core needle biopsy removes more than 1 tissue sample, but usually through the same area on the breast. The samples are small. The needle is often guided into the tumor with imaging

One or more clips may be placed near the breast tumor during a biopsy. The clips are small, painless, and made of metal. They mark the site for future treatment and imaging. The clips cause no problems, even if they're left in place for a long time. You'll be able to go through airport security and have an MRI.

Biopsy of metastasis

Metastasis is the spread of cancer to an area of the body, such as bone, lungs, or liver. A biopsy of the metastasis is typically needed to confirm the presence of cancer. If there's more than 1 metastasis, more than 1 site may be biopsied. The type of biopsy used depends on the location of the suspected metastases and other factors. These biopsies are usually guided by imaging tests like ultrasound or CT scan.

Biopsy results

Your pathology report will contain information about tumor histology and grade (how abnormal the tumor looks). Histology is the study of the anatomy (structure) of cells, tissues, and organs under a microscope. It's used to make treatment decisions.

You may be recommended to have an open biopsy (surgery) to remove (excise) the tumor to confirm histology. Talk to your health care provider for more information on next steps.

HER2 status

Breast cancer can produce greater than normal amounts of human epidermal growth factor receptor 2 (HER2). HER2 is a protein involved in normal cell growth. It's found on the surface of all cells. When amounts are high, it causes cells to grow and divide. Some breast cancers have too many HER2 genes or receptors. This is called HER2-positive (HER2+) breast cancer. You might hear it called HER2 overexpression or amplification.

There are 2 tests for HER2:

- **Immunohistochemistry (IHC)** measures receptors. If the score is 0 or 1, it's considered HER2-negative (HER2-). If the score is 2+, further testing is needed. If the IHC score is 3+, the cancer is considered HER2-positive (HER2+). Additionally, another term you may hear is HER2-low, which means HER2 IHC1+ or 2+ but FISH-negative.
- **Fluorescence in situ hybridization (FISH)** counts the number of copies of the *HER2* gene. This test is done mainly when the IHC score is unclear.

HER2 testing should be done on all new tumors. A tumor biopsy sample will be used. You might have more than one HER2 test.

Newer drugs that target HER2 have recently been shown to be effective in some people who have metastatic breast cancer with lower levels of HER2 expression (0, 1+, and 2+ by IHC but FISH is negative) called HER2-low or HER2-ultralow.

Immunohistochemistry

Immunohistochemistry (IHC) is a special staining process that involves adding a chemical marker to cancer or immune cells. The cells are then studied using a microscope. IHC can find estrogen, progesterone, and HER2 receptors in breast cancer cells. A pathologist will measure the number of cells with estrogen or progesterone receptors and the number of receptors inside each cell.

FISH or ISH

FISH, or other ISH methods like dual ISH, are testing methods that involve special dyes, called probes, that attach to pieces of DNA. DNA is the genetic material in a person's cells.

Hormone receptor status

Estrogen and progesterone are hormones. A hormone is a substance made by a gland in your body. Hormones attach or bind to receptors found on the surface or inside certain cells.

When estrogen and progesterone attach to receptors found inside breast cancer cells, they can cause cancer to grow. Hormone receptor testing looks for estrogen and progesterone receptors in breast cancer cells. If found, these receptors may be targeted using endocrine (hormone-blocking) therapy.

Breast cancers can be hormone receptor-positive (HR+) or hormone receptor-negative (HR-). Hormone receptor testing should be done on any new tumors. A biopsy sample will be used.

Hormone receptor-positive

In hormone receptor-positive (HR+), or hormone-sensitive, breast cancer, IHC finds estrogen receptors (ER+), progesterone receptors (PR+), or both (ER+/PR+). Most breast cancers are ER+/PR+ or ER+/PR-. An ER-/PR+ tumor is relatively uncommon.

Hormone receptor-negative

Hormone receptor-negative (HR-) breast cancer cells don't express estrogen or progesterone hormone receptors. HR- cancers often grow faster than HR+ cancers. Both the estrogen and progesterone receptors need to be negative for breast cancer to be considered HR-.

Estrogen receptor-positive (ER+) breast cancer cells

- ✓ In ER+ breast cancer, testing finds estrogen hormone receptors in at least 1 out of every 100 cancer cells.
- ✓ In ER-low–positive breast cancer, testing finds estrogen hormone receptors in 1 to 10 out of every 100 cancer cells.

Biomarker testing

Biomarker testing includes tests of genes or their products (proteins). It identifies the presence or absence of mutations and certain proteins that might suggest treatment. Some biomarker tests are described next.

Tumor mutation testing

A sample of your tumor or blood may be used to see if the cancer cells have any specific DNA mutations. This testing differs from the genetic testing for mutations you may have inherited from your birth parents. In tumor mutation testing, only the tumor is tested.

Testing is done using a variety of methods, such as FISH, ISH, IHC, next-generation sequencing (NGS), and/or polymerase chain reaction (PCR). These methods are used to identify the presence of gene mutations, alterations, rearrangements, or fusions found in the tumor or metastasis.

- Certain mutations, such as *PIK3CA*, *AKT1*, *PTEN*, *ESR1*, *NTRK*, and *RET*, can be targeted with specific therapies.
- Testing for *ESR1* and *RET* mutations is done on hormone receptor-positive (HR+) tumors.
- Germline *BRCA1/2* and *PALB2* testing is recommended for everyone with metastatic breast cancer, unless it's already been done (more recently than 2014). See "Genetic cancer risk testing" on page 17 for more information.

PD-1 and PD-L1 testing

Programmed cell death protein 1 (PD-1) and programmed death-ligand 1 (PD-L1) are immune proteins. If either protein is expressed on the surface of cancer cells, it can cause your immune cells to ignore the cancer and suppress the anti-tumor immune response. If your cancer expresses either protein, you might have treatment that combines chemotherapy and immune checkpoint inhibitors. This is designed to activate your immune system to better fight the cancer cells. Usually, this test is done when hormone receptors and HER2 testing are negative (like in triple-negative tumors).

Tumor mutational burden

When there are 10 or more mutations per million base pairs of tumor DNA, it's called tumor mutational burden-high (TMB-H). TMB-H can be used to help predict response to cancer treatment using immune checkpoint inhibitors that target the proteins PD-1 and PD-L1.

Testing takes time. It might take weeks for all test results to come in.

MSI-H/dMMR mutation

Microsatellites are short, repeated strings of DNA. When errors or defects occur, they're fixed by mismatch repair (MMR) proteins. Some cancers have DNA mutations for changes that prevent these errors from being fixed. This is called microsatellite instability (MSI) or MMR deficient (dMMR). When cancer cells have more than a normal number of microsatellites, it's called MSI-high (MSI-H). This is often due to dMMR genes.

Tumor markers

Your blood or biopsy tissue may be tested for proteins called tumor markers. An increase in the level of certain tumor markers could mean the cancer has grown or spread more (progressed). However, not everyone has elevated levels of these markers, and tumor markers alone aren't a reliable method of detecting breast cancer. Therefore, they aren't routinely checked and depend on your individual situation.

Liquid biopsy

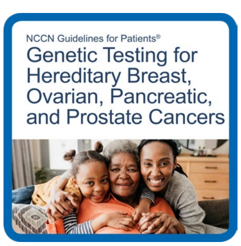
Some abnormal changes (mutations) can be found by testing circulating tumor DNA (ctDNA) in the blood. In a liquid biopsy, a sample of blood is taken to look for cancer cells or for pieces of DNA from tumor cells. Sometimes, testing can quickly use up a tumor sample, and a liquid biopsy might be an option in this case.

Genetic cancer risk testing

About 1 out of 20 breast cancers are hereditary, which means they are passed down from birth parents through DNA. Depending on your family history, age, or other features of your cancer, your health care provider might refer you for genetic counseling and hereditary genetic testing to learn more about your cancer. A genetic counselor or trained provider will speak with you about the results. Test results may guide treatment planning.

Genetic testing is done using blood or saliva. The goal is to look for gene mutations inherited from your biological (birth) parents, called germline mutations. Some mutations can put you at risk of more than 1 type of cancer. You can pass these genes on to your children. Also, other family members might carry these mutations. Tell your care team if there's a family history of cancer.

More information on genetic cancer risk testing can be found in N*CCN Guidelines for Patients: Genetic Testing for Hereditary Breast, Ovarian, Pancreatic, and Prostate Cancers*, available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.



BRCA mutation tests

Everyone has *BRCA* genes. Normal *BRCA* genes help prevent tumor growth, help fix damaged cells, and help cells grow normally. However, mutations in *BRCA1* or *BRCA2* increase the risk of breast, ovarian, prostate, colorectal, pancreatic, and melanoma skin cancers. Mutated *BRCA* genes can also affect how some treatments work.

The technology for genetic testing has improved, so if you tested negative for a gene mutation over 10 years ago, you may be re-tested.

Other genes that might be tested

Other genes, such as *PALB2*, *p53*, *CHEK2*, and *ATM*, might be tested. For example, *PALB2* normally helps prevent cancer. When *PALB2* mutates, it no longer works correctly. People with a *PALB2* mutation have a higher risk of developing breast cancer.

Key points

- Not all breast cancers are the same. Treatment planning starts with testing. Results from a tumor biopsy and imaging studies are used to determine what treatments are best for your type of breast cancer.
- During a biopsy, tissue or fluid samples are removed for testing. Samples are needed to confirm the presence of cancer and to perform cancer cell tests.
- A sample from a biopsy of your tumor will be tested for estrogen receptor (ER) status, progesterone receptor (PR) status, HER2 status, tumor mutation, PD-L1, and histology. This provides information about the behavior of your cancer, as well as treatments to which your cancer may respond. Other biomarker tests may be done.
- About 1 out of 20 breast cancers are hereditary. Depending on your family history, age, or other features of your cancer, your health care provider might refer you for hereditary genetic testing or to a genetic counselor.

Questions to ask

- What type(s) of imaging scans will I have?
- What areas will be biopsied?
- What tests will be done on the tumor or metastasis?
- When will the test results be ready and who will discuss them with me?
- What's the tumor HER2 and hormone receptor (HR) status?
- Does the tumor have any unique features or biomakers, such as PD-L1, that might suggest treatment?

What's next?

The next chapter provides information on the types of treatment and what to expect from treatment.

share with us.

Take our survey and help make the NCCN Guidelines for Patients better for everyone!

NCCN.org/patients/comments

3

Types of treatment

This chapter provides a general overview of the types of systemic therapy used to treat metastatic breast cancer. Systemic therapy is drug therapy that works throughout the body. It's the main treatment for metastatic breast cancer.

Your care team

Treating someone with metastatic breast cancer takes a team approach. Treatment decisions should involve a team of health care and psychosocial care professionals from different backgrounds who have knowledge and experience in your type of cancer. This team is united in planning and implementing your treatment.

Systemic therapy

Systemic therapy is a type of cancer treatment that travels in the bloodstream to kill cancer cells throughout the body. It's the main treatment for metastatic breast cancer. Types of systemic therapy include endocrine (hormone) therapy, chemotherapy, targeted therapy, and immunotherapy. The goal of treatment is to slow the spread of cancer.

Treatment for metastatic breast cancer is a combination of systemic therapies that are often given in a specific order or sequence (called sequential). The choice of therapy takes into consideration many factors, including hormone receptor and HER2 tumor status, your preferences about treatment, and if you have any other serious health conditions. The goals of systemic therapy should be discussed before starting treatment.

All systemic treatments listed in this guide are recommended and appropriate. When helpful, NCCN experts also assign a level of preference to their recommendations for systemic therapies:

- **Preferred therapies** have the most evidence they may work better and be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

For a general list of systemic therapies, see **Guide 2**.

Guide 2
Systemic therapy examples (listed in alphabetical order)

Chemotherapy examples	• Capecitabine (Xeloda) • Carboplatin (Kyxata) • Cisplatin • Cyclophosphamide (Frindovyx) • Docetaxel • Doxorubicin (Doxi) • Epirubicin (Ellence) • Eribulin (Halaven) • Fluorouracil • Gemcitabine (Avgemsi) • Methotrexate • Paclitaxel • Vinorelbine
Antibody-drug conjugate (ADC) examples	• Ado-trastuzumab emtansine (T-DM1) (Kadcyla) • Datopotamab deruxtecan-dlnk (Datroway) • Fam-trastuzumab deruxtecan-nxki (T-DXd) (Enhertu) • Sacituzumab govitecan-hziy (Trodelvy)
Targeted therapy examples (these don't target HER2)	CDK4/6 inhibitors • Abemaciclib (Verzenio), Palbociclib (Ibrance), and Ribociclib (Kisqali)
	PARP inhibitors • Olaparib (Lynparza) and Talazoparib (Talzenna)
	PIK3CA, AKT1, PTEN, ROS1, TRK, mTOR, and other inhibitors • Alpelisib (Piqray), Capivasertib (Truqap), Entrectinib (Entrectinib), Erdafitinib (Balversa), Everolimus (Afinitor), Inavolisib (Itovebi), Larotrectinib (Vitrakvi), Repotrectinib (Augtyro), and Selpercatinib (Retevmo)
HER2-targeting therapy examples	Antibodies • Margetuximab-cmkb (Margenza) • Pertuzumab (Perjeta) and its substitutes called biosimilars • Trastuzumab (Herceptin) and biosimilars
	ADCs • Ado-trastuzumab emtansine and Fam-trastuzumab deruxtecan-nxki
	Inhibitors • Lapatinib (Tykerb), Neratinib (Nerlynx), and Tucatinib (Tukysa)
Immunotherapy	• Pembrolizumab (Keytruda) and Dostarlimab-gxly (Jemperli)
Endocrine therapy	• Endocrine therapy can be found in **Guide 3.**

Endocrine therapy

Endocrine therapy blocks estrogen or progesterone to treat hormone receptor-positive (HR+) breast cancer. The endocrine system is made up of organs and tissues that produce hormones. Hormones are natural chemicals released into the bloodstream.

There are 4 hormones that might be targeted in endocrine therapy:

- **Estrogen** is made mainly by the ovaries but is also made by other tissues in the body, such as fat tissue.
- **Progesterone** is made mainly by the ovaries.
- **Luteinizing hormone-releasing hormone (LHRH)** is made by a part of the brain called the hypothalamus. It tells the ovaries to make estrogen and progesterone and the testicles to make testosterone. LHRH is also called gonadotropin-releasing hormone (GnRH).
- **Androgen**, such as testosterone, is made by the adrenal glands, testicles, and ovaries.

Hormones may cause breast cancer to grow. Endocrine therapy stops your body from making hormones or blocks what hormones do in the body. This can slow tumor growth or shrink the tumor for a period of time.

Endocrine therapy isn't the same as hormone replacement therapy (HRT) used for menopause.

In addition to commonly used aromatase inhibitors, another class of drugs, such as oral SERDs (selective estrogen receptor degraders), is emerging as a major treatment option for breast cancer tumors with an *ESR1* mutation.

Types of endocrine therapy can be found in **Guide 3**.

Testosterone

For people assigned male at birth whose bodies continue to make testosterone, endocrine therapy includes tamoxifen or an aromatase inhibitor with a testosterone-suppressing therapy.

Premenopause

If you have menstrual periods or recently stopped menstruating but still have elevated blood estrogen levels, you're in premenopause. In premenopause, the ovaries are the main source of estrogen and progesterone. GnRH agonists may be used to temporarily induce menopause for people in premenopause. A combination of GnRH agonists and tamoxifen or aromatase inhibitors (AIs) may be considered as endocrine therapy for people in premenopause.

Menopause

In menopause, the ovaries permanently stop producing hormones and menstrual periods stop. Estrogen and progesterone levels are low, but the adrenal glands, liver, and body fat continue to make small amounts of estrogen. If you don't have periods, a test using a blood sample may be used to confirm your status.

Cancer treatment can cause temporary menopause. If you stopped having periods due to removal of your uterus (hysterectomy) but you still have your ovaries, then your menopausal status is confirmed with a blood test. If both ovaries have been removed (with or without your uterus), you're in menopause.

Preventing pregnancy during treatment

If you become pregnant during endocrine therapy, it can cause birth defects. Non-hormonal birth control methods, such as intrauterine devices (IUDs) and barrier methods, are preferred in people with a breast cancer diagnosis. Types of barrier methods include condoms, diaphragms, cervical caps, and contraceptive sponges.

Guide 3
Endocrine therapy types

Bilateral oophorectomy	Surgery to remove both ovaries
Ovarian ablation	Radiation to permanently stop the ovaries from making hormones
Ovarian or testosterone suppression	Drugs to temporarily stop the ovaries or testicles from making hormones, such as LHRH and GnRH • LHRH agonists include Goserelin (Zoladex) and Leuprolide (Lupron Depot). These are injected every 4 or 12 weeks. They don't affect estrogen made by the ovaries • GnRH agonists might be used to suppress ovarian hormone or testosterone production
Aromatase inhibitors (AIs)	Drugs to stop a type of hormone called androgen from changing into estrogen by interfering with an enzyme called aromatase. AIs don't affect estrogen made by the ovaries. Nonsteroidal AIs include Anastrozole (Arimidex) and Letrozole (Femara). Exemestane (Aromasin) is a steroidal AI
Estrogen receptor (ER) modulators or anti-estrogens	• Selective estrogen receptor modulators (SERMs) block estrogen from attaching to hormone receptors. Tamoxifen and Toremifene (Fareston) are SERMs • Selective estrogen receptor degraders (SERDs) block and destroy estrogen receptors. Elacestrant (Orserdu), Fulvestrant (Faslodex), Imlunestrant (Inluriyo), and Vepdegestrant (Veppanu) are SERDs
Hormones	Examples include Estradiol, Fluoxymesterone, and Megestrol acetate (Megace)

Chemotherapy

Chemotherapy kills fast-dividing cells throughout the body, including cancer cells and some normal cells. More than 1 chemotherapy may be used to treat breast cancer. When only 1 drug is used, it's called a single agent. A combination or multi-agent regimen is the use of 2 or more drugs.

Some chemotherapy drugs are liquids infused into a vein or injected under the skin with a needle. Other chemotherapy drugs may be given as a pill.

Some examples of chemotherapy drugs include the following:

- **Anthracyclines** include doxorubicin (Adriamycin), doxorubicin liposomal injection (Doxil), and epirubicin (Ellence)
- **Taxanes** include docetaxel, paclitaxel, and albumin-bound paclitaxel.
- **Antimetabolites** include capecitabine (Xeloda), fluorouracil, gemcitabine, and methotrexate.

Most chemotherapy is given in cycles of days of treatment followed by days of rest. This allows the body to recover before the next cycle. Cycles vary in length depending on which drugs are used. The number of treatment days per cycle and the total number of cycles also vary.

Warnings about supplements and drug interactions

Some supplements and foods can affect the ability of a drug used to treat breast cancer to do its job. This is called a drug interaction. Since there's often a lack of published information about how certain supplements interact with standard medicines, your care team may recommend that many of them be discontinued during treatment.

Also, certain medicines can affect the ability of a drug to do its job. Antacids, heart or blood pressure medicine, and antidepressants are just some of the medicines that might interact with systemic therapy or supportive care medicines given during systemic drug therapy.

Therefore, it's very important to tell your care team about all medicines, vitamins, over-the-counter drugs, herbals, or supplements you're taking.

Bring a list with you to every visit.

Antibody-drug conjugates

An antibody-drug conjugate (ADC) delivers cell-specific chemotherapy. It attaches to a protein found on the outside of the cancer cell, then enters the cell. Once inside the cell, chemotherapy is released. ADCs are given in cycles. Some ADCs target HER2.

ADCs include ado-trastuzumab emtansine, datopotamab deruxtecan-dlnk, fam-trastuzumab deruxtecan-nxki, sacituzumab govitecan-hziy.

HER2-targeted therapy

When breast cancer has higher amounts of HER2, it's called HER2-positive (HER2+) breast cancer.

HER2-targeted therapy treats HER2+ breast cancer. It can be given alone, with chemotherapy, or with endocrine therapy.

HER2-targeted therapies include:

- **HER2 antibodies**, such as trastuzumab, pertuzumab, and margetuximab, prevent HER2 growth signals from outside the cell. They also help immune cells attack cancer cells.
- **HER2 inhibitors**, such as tucatinib, lapatinib, and neratinib, stop HER2 growth signals from within the cell.
- **HER2 conjugates or HER2 antibody-drug conjugates (ADCs)** deliver cell-specific chemotherapy. They attach directly to HER2s then enter the cell. Once inside, chemotherapy is released.

Treatment options by cell receptor type

There are many treatments for breast cancer. Which ones are right for you are based on many factors. **Two important factors are the hormone receptor (HR) and HER2 status of any tumors.**

- ✓ Hormone receptors include estrogen and progesterone. A tumor is considered hormone receptor-positive (HR+) if an increased number of estrogen receptors, progesterone receptors, or both are found.
- ✓ HER2 is a protein involved in normal cell growth. There might be higher amounts of HER2 in your breast cancer. HER2 is categorized as negative, 0+, low, or positive.

Endocrine therapy stops cancer growth caused by hormones. It's a standard treatment for hormone receptor-positive (HR+) cancers. HR+ cancer can be estrogen receptor-positive (ER+), progesterone receptor-positive (PR+), or both (ER+/PR+).

HER2-targeted therapy is a standard treatment for HER2-positive (HER2+) cancers.

Other systemic therapies are used with those listed above to treat metastatic breast cancer.

Your heart will be monitored before and during treatment with HER2-targeted therapy. Tests will measure the left ventricular ejection fraction, the amount of blood pumping from the left side of the heart.

Other targeted therapies

This section discusses inhibitors that are different from inhibitors used in HER2-targeted therapy.

CDK4/6 inhibitors

CDK is a cell protein that helps cells grow and divide. For hormone receptor-positive (HR+), HER2-negative (HER2-) cancer, taking a CDK4/6 inhibitor with endocrine therapy may help control cancer longer and improve survival. CDK4/6 inhibitors include abemaciclib, palbociclib, and ribociclib.

PARP inhibitors

Cancer cells often become damaged. PARP is a cell protein that repairs cancer cells and allows them to survive. Blocking PARP can cause cancer cells to die. Olaparib and talazoparib are examples of PARP inhibitors.

PIK3CA, PTEN, AKT1, and other inhibitors

The *PIK3CA* gene is one of the most frequently mutated genes in breast cancers. *PTEN* and *AKT1* can also be altered in breast cancers. A mutation or alteration in these genes can lead to increased growth of cancer cells and resistance to various treatments. Alpelisib and inavolisib are examples of a PIK3CA inhibitor and capivasertib is an AKT1 inhibitor.

mTOR inhibitors

mTOR is a cell protein that helps cells grow and divide. Endocrine therapy may stop working if mTOR becomes overactive. mTOR inhibitors help endocrine therapy work again.

Everolimus (Afinitor) is an mTOR inhibitor. It can be taken with exemestane, fulvestrant, or tamoxifen.

Immunotherapy

Immunotherapy tries to reactivate the immune system against tumor cells. The immune system has many on and off switches. Tumors take advantage of off switches. Immunotherapy can block these off switches, which helps the immune system turn on. Immunotherapy can be given alone or with other types of treatment. Immunotherapy is effective in metastatic breast cancer if the tumor tests positive for PD-L1 or high tumor mutational burden (TMB).

Bone-strengthening therapy

Medicines that target the bones may be given to help relieve bone pain or reduce the risk of bone-related problems. Some medicines work by slowing or stopping bone breakdown, while others help increase bone thickness.

Breast cancer can spread (metastasize) to your bones. This puts your bones at risk of injury and disease. Some treatments for breast cancer, like aromatase inhibitors or GnRH agonists, can cause bone loss (osteoporosis), which puts you at an increased risk of fractures.

Drugs used to prevent bone loss and fractures:

- Oral bisphosphonates
- Zoledronic acid (Zometa)
- Pamidronate (Aredia)
- Denosumab (Prolia)

Drugs used to treat bone metastases:

- Zoledronic acid (Zometa)
- Pamidronate (Aredia)
- Denosumab (Xgeva)

You may be screened for bone weakness (osteoporosis) using a bone mineral density test. This measures how much calcium and other minerals are in your bones. It's also called a dual-energy x-ray absorptiometry (DEXA) scan and is painless. Bone mineral density tests look for osteoporosis and help predict your risk of bone fractures.

Zoledronic acid, pamidronate, and denosumab

Zoledronic acid, pamidronate, and denosumab are used to prevent bone loss (osteoporosis) and fractures. Zoledronic acid and denosumab are also used in people with breast cancer who have bone metastases to help reduce

Standard of care is the best-known way to treat a particular disease based on past clinical trials. There may be more than one treatment regimen that is considered standard of care. Ask your care team what treatment options are available and if a clinical trial might be right for you.

the likelihood of fractures, pain, or other complications.

You might have blood tests to monitor kidney function, calcium levels, and magnesium levels. A calcium and vitamin D supplement will likely be recommended by your doctor.

Let your dentist know if you're taking any of these medicines. Also, ask your care team how these medicines might affect your teeth and jaw. Osteonecrosis, or bone tissue death, of the jaw is a rare but serious side effect. Tell your care team about any planned trips to the dentist and surgeries or dental procedures that might also affect the jawbone. It's important to take care of your teeth and to see a dentist regularly.

Clinical trials

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of treating cancer need to be studied in people. If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It is common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There's no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.

Finding a clinical trial

In the United States

NCCN Cancer Centers
NCCN.org/cancercenters

The National Cancer Institute (NCI)
cancer.gov/about-cancer/treatment/clinical-trials/search

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
cancer.gov/contact

Key points

- Systemic therapy works throughout the body and is the main treatment for metastatic breast cancer.
- Systemic treatment is based on tumor estrogen receptor (ER), progesterone receptor (PR), and HER2 expression.
- Treatment is a combination of systemic therapies that are often given in a specific order or sequence (called sequential).
- HER2-targeted therapy is drug therapy that treats HER2+ breast cancer.
- Endocrine therapy stops your body from making hormones or blocks what hormones do in the body. It's used to treat hormone receptor-positive (HR+) breast cancer.
- Medicines that target the bones may be given to help relieve bone pain or reduce the risk of bone-related problems.
- A clinical trial is a type of research that studies a treatment to see how safe it is and how well it works.

Questions to ask

- What is your experience treating breast cancer?
- What combination of systemic therapies do you recommend and why?
- Who will be a part of my care team?
- What websites about my condition do you recommend? Is there any information I can take home to read?
- Am I a candidate for a clinical trial?

What's next?

Now that you've read about the different types of systemic therapy and what to expect, the next chapter talks about some general side effects of treatment and what might be done to manage those effects.

4
Supportive care

Supportive care helps manage the symptoms of metastatic breast cancer and the side effects of treatment. This chapter discusses possible treatment side effects.

People are living longer with metastatic breast cancer. Since treatment can last many years, supportive care for side effects is important.

What is supportive care?

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care is for everyone with cancer and their families, not just for those at the end of life.

Supportive care includes a wide range of services. Supportive care prevents or manages the symptoms of cancer and the side effects of cancer treatment, like pain and cancer-related fatigue. It also addresses the mental, social, emotional, and spiritual concerns faced by people with cancer.

Supportive care provides help with additional needs, such as:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advance care and end of life

Read more about the types of support you may receive in *NCCN Guidelines for Patients: Palliative Care*, available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

“When you are going through treatment that’s difficult, remember it’s about the cancer. Don’t let the side effects of treatment become your focus!”

Side effects

All cancer treatments can cause unwanted health issues called side effects. Side effects depend on many factors. These factors include the drug type and dose, length of treatment, and the person. Some side effects may just be unpleasant. Others are very serious. Tell your care team about any new or worsening symptoms.

Some common side effects are described next.

Blood clots

Cancer treatment can cause blood clots to form. This can block blood flow and oxygen delivery to parts of the body. Blood clots can break loose and travel to other parts of the body, causing breathing problems, stroke, or other health issues.

Bone health

Breast cancer may spread to your bones. Some breast cancer treatments may also weaken your bones. Both can put your bones at increased risk of injury and disease. Such problems include fractures, bone pain, and squeezing (compression) of the spinal cord. High levels of calcium in the blood, called hypercalcemia, may also occur.

Medicine may be given to help relieve bone pain and reduce the risk of other bone problems. Some medicines work by slowing or stopping bone breakdown, while others help increase bone thickness. Taking calcium and vitamin D with these bone health medicines is recommended, but talk to your care team first.

Diarrhea or constipation

Diarrhea is frequent and watery bowel movements. Your care team will tell you how to manage diarrhea. When it occurs, it's important to drink lots of fluids.

Constipation means having less frequent and more difficult bowel movements. Constipation is also common, especially if you're taking certain pain medicines and some anti-nausea medicines. Drinking fluids, staying active, and taking medicines for constipation are often recommended.

Emotional distress

Depression, anxiety, and sleeping issues are common during a cancer diagnosis. Talk to your care team and those with whom you feel most comfortable about how you're feeling. There are services, people, and medicine that may be recommended to help relieve your distress.

Fatigue

Fatigue is extreme tiredness and inability to function due to a lack of energy. Fatigue may be caused by cancer or it may be a side effect of treatment. Let your care team know how you're feeling and if fatigue is getting in the way of doing the things you enjoy. Eating a balanced diet, exercise, yoga, acupuncture, and massage therapy can help. You might be referred to a nutritionist or dietitian to help with fatigue.

Hair loss

Chemotherapy may cause hair loss (alopecia) all over your body—not just on your scalp. Some chemotherapy drugs are more likely than others to cause hair loss. Dosage might also affect the amount of hair loss. Most of the time, hair loss from chemotherapy is temporary. Hair often regrows 3 to 6 months after treatment ends. Your hair may be a different shade or texture at first. Scalp cooling (or scalp hypothermia) might help lessen hair loss in people receiving certain types of chemotherapy.

Infections

Infections are more common and more severe in people with a weakened immune system. Drug treatment for metastatic breast cancer can lower the body's natural defense against infections. If not treated early, infections can be fatal.

Neutropenia, a low number of white blood cells that often occurs during treatment, can lead to frequent or severe infections. When someone with neutropenia develops a fever, it's called febrile neutropenia (FN). A fever is a temperature of over 100.4º F. With FN, your risk of infection may be higher than normal. This is because a low number of white blood cells leads to a reduced ability to fight infections. Sometimes the infection can be from bacteria entering the bloodstream, which is very dangerous. FN is a side effect of some types of systemic therapy. Ask your care team how to prevent FN, what to look for, and what to do in an emergency.

Interstitial lung disease

Interstitial lung disease, also called ILD or pneumonitis, describes a large group of conditions that cause inflammation and scarring of lung tissue. Symptoms include shortness of breath, cough, fatigue, or tightness in the chest. ILD is a posssible side effect of some systemic therapies.

Loss of appetite

The side effects from cancer or its treatment and the stress of having cancer might cause you to feel not hungry or sick to your stomach (nauseated). You might have a sore mouth or difficulty swallowing.

Healthy eating is important during treatment. It includes eating a balanced diet, eating the right amount of food, and drinking enough fluids. A registered dietitian who's an expert in nutrition and food can help.

Low blood cell counts

Some cancer treatments can cause low blood cell counts.

- **Anemia** is a condition where your body doesn't have enough healthy red blood cells, resulting in less oxygen being carried to your organs. You might tire easily if you're anemic.
- **Neutropenia** is a decrease in neutrophils, a type of white blood cell. This puts you at risk of infection.
- **Thrombocytopenia** is a condition where there aren't enough platelets found in the blood. This puts you at risk of bleeding.

Lymphedema

Lymphedema is a condition in which lymph fluid builds up in tissues and causes swelling. Lymph node metastases can sometimes cause lymphedema. If you have lymphedema, you may be referred to an expert in lymphedema management.

Nausea and vomiting

Nausea and vomiting are common side effects of treatment. You may be given medicine to prevent nausea and vomiting.

Neuropathy and neurotoxicity

Some treatments can damage the nervous system (neurotoxicity), causing neuropathy and problems with concentration, memory, and thinking. Neuropathy is a nerve problem that causes pain, numbness, tingling, swelling, or muscle weakness in different parts of the body. It usually begins in the hands or feet and gets worse with additional cycles of treatment.

Pain

Tell your care team about any pain or discomfort. You might meet with a palliative care specialist or with a pain specialist to manage pain

Palliative care

Palliative care is appropriate for anyone, regardless of age, cancer stage, or the need for other therapies. It focuses on physical, emotional, social, and spiritual needs that affect quality of life.

Late effects

Late effects are side effects that occur months or years after a disease is diagnosed or after treatment has ended. Late effects may be caused by cancer or cancer treatment. They may include physical, mental, and social health concerns, as well as second cancers. The sooner late effects are treated, the better. Ask your care team about what late effects could occur. This will help you know what to look for.

Supportive care resources

More information on supportive care is available at [NCCN.org/patientguidelines] and on the [NCCN Patient Guides for Cancer] app.

Key points

- Supportive care is health care that relieves symptoms caused by treatment and improves quality of life. Supportive care is always given.
- All cancer treatments can cause unwanted health issues called side effects. Side effects depend on many factors. These factors include the drug type and dose, length of treatment, and the person.
- Some side effects are very rare. Ask your care team what to expect.
- Tell your care team about any new or worsening symptoms.
- Late effects are side effects that occur months or years after a disease is diagnosed. Late effects may include physical, mental, and social health concerns, and second cancers.
- People living with metastatic breast cancer often have unique challenges not experienced by other breast cancer survivors. Your cancer care team or one of your primary health care providers can be a great source of information and support if you're experiencing any changes in your mental, physical, emotional, sexual, or financial health.

Questions to ask

- Who will coordinate my care?
- How long should I wait if I notice changes in my condition and who should I call?
- What should I do on weekends and other non-office hours?
- Will my care team be able to communicate with the emergency department or urgent care team?
- Are there resources to help me pay for treatment or other care I may need?

What's next?

The next chapter provides more detailed information on your specific treatment options based on the tumor hormone receptor (HR) and HER2 status, and any biomarkers that might be found.

5

Your treatment options

Metastatic breast cancer is treated with systemic (drug) therapy. The choice of therapy is based on the tumor hormone receptor (HR) and HER2 status, any biomarkers that might be found, your overall health, and shared decision-making between you and your care team. This chapter provides detailed, specific treatment options. Together, you and your care team will choose a treatment plan that's best for you.

Treatment for metastatic breast cancer includes multiple rounds of systemic (drug) therapy that can last many years. Treatment is given in a specific order or sequence, called sequential therapy. It starts with first-line therapy, followed by second-line therapy, and so on. The choice of therapy is based on the hormone receptor (HR) and HER2 status of the tumor, as well as any biomarkers that might be found. The following pages divide treatment by HR and HER2 status.

Your goals of systemic therapy should be discussed throughout treatment. It's important to have regular talks with your care team about any side effects or changes in your condition.

HR+ and HER2- metastatic breast cancer

In hormone receptor-positive (HR+) breast cancer, estrogen (ER+), progesterone receptors (PR+), or both (ER+/PR+) are found. In HER2-negative (HER2-) breast cancer, tumor cells don't express increased levels of HER2.

"I was so fortunate to find an expert health care team that I could talk to openly and honestly during and after my treatment. This shared decision-making made me feel like I had a say about my care."

Initial treatment for HR+, HER2- metastatic breast cancer most often starts with endocrine therapy plus a CDK4/6 inhibitor (abemaciclib, ribociclib, or palbociclib). The next treatment depends on the tumor's response to the initial treatment, the types of mutations specific to your cancer, and if your situation or health has changed.

Typically, the most effective, least toxic therapies are chosen first. Therefore, endocrine-based therapies are usually given before chemotherapy-based therapies.

Visceral crisis

When cancer within internal organs (metastasis) causes severe symptoms or causes organs to stop working as they should, it's called a visceral crisis. If you're in visceral crisis, the goal is to decrease the amount of cancer quickly. This may be accomplished by using chemotherapy or other systemic therapies.

No visceral crisis

If you aren't in visceral crisis, treatment starts with endocrine therapy, often combined with CDK4/6 inhibitors. CDK4/6 inhibitors include abemaciclib, palbociclib, and ribociclib. Later, systemic therapy specific to HER2- breast cancer will be given. People who are premenopausal might have ovarian suppression or ablation in addition to endocrine therapy. Systemic therapy options (including endocrine therapy) can be found in **Guide 4**.

Systemic therapy is based on the cancer's hormone receptor (HR) and HER2 status, and any biomarkers that might be found.

Sometimes, HR+ cancer doesn't respond or stops responding to endocrine therapy. If this happens, chemotherapy or antibody-drug conjugates (ADCs) are used. If an ER+ tumor is resistant to endocrine therapy (called endocrine refractory), then systemic therapy options found under triple-negative breast cancer on page 42 will be considered.

HR+ and HER2+ metastatic breast cancer

Breast cancer that's HR+ and HER2+ is sometimes called triple-positive breast cancer. In triple-positive breast cancer, tumor cells have estrogen receptors (ER+), progesterone receptors (PR+), and a larger than normal number of HER2 receptors on their surface. It's treated with a combination of HER2-targeted therapy, other systemic therapies that may include chemotherapy, antibody-drug conjugates, targeted therapy, or endocrine therapy. Endocrine therapy can include aromatase inhibitors (AIs), tamoxifen, fulvestrant, and CDK 4/6 inhibitors. People who are premenopausal might have ovarian suppression or ablation in addition to endocrine therapy.

Guide 4
Systemic therapy options for HR+ and HER2- without visceral crisis

Preferred first-line therapies	• Aromatase inhibitor plus CDK4/6 inhibitor (Ribociclib, Abemociclib, or Palbociclib) • For *PIK3CA* mutation, Fulvestrant with Inavolisib and Palbociclib or Alpelisib with Fulvestrant • Fulvestrant with CDK4/6 inhibitor • For germline *BRCA1* or *BRCA2* mutations, Olaparib or Talazoparib • Chemotherapy, antibody-drug conjugates, or targeted therapy
Preferred second-line and next-line therapies	• Fulvestrant plus CDK4/6 inhibitor if CDK4/6 inhibitor not used before • For *PIK3CA, AKT1,* or *PTEN* mutation, Capivasertib with Fulvestrant • Everolimus plus endocrine therapy (Exemestane, Fulvestrant, or Tamoxifen)
Other recommended therapies	• For an *ESR1* mutation after prior endocrine therapy with or without a CDK4/6 inhibitor, Elacestrant, Imlunestrant, or Vepdegestrant • Fulvestrant plus Anastrozole or Letrozole • Fulvestrant or Anastrozole or Letrozole or Tamoxifen or Exemestane • For germline *PALB2,* Olaparib • Chemotherapy or targeted therapy
Therapies used in certain cases	• Megestrol acetate • Estradiol • Abemaciclib • Imlunestrant plus Abemaciclib • Targeted therapies • For *NTRK* gene fusion, Entrectinib, Larotrectinib, or Repotrectinib • For MSI-H/dMMR, Pembrolizumab or Dostarlimab-gxly • For TMB-H, Pembrolizumab • For *RET* gene fusion, Selpercatinib • For somatic *BRCA1* or *BRCA2* mutations, Olaparib • For *FGFR1-3* fusions or mutations, Erdafitinib • For HER2 activating mutations, Fulvestrant and Neratinib plus Trastuzumab or Tucatinib plus Trastuzumab with or without Fulvestrant or Neratinib with or without Fulvestrant
Notes	• People who are in premenopause or perimenopause might have ovarian suppression or ablation in addition to endocrine therapy • Targeted therapy might be used if specific mutations are found

HER2-targeted therapy and other systemic therapies are found in **Guide 5**.

HR- and HER2+ metastatic breast cancer

In hormone receptor-negative (HR-) breast cancer, cancer cells test negative for estrogen (ER-) and progesterone receptors (PR-). In HER2-positive (HER2+) breast cancer, HER2 receptors are found (amplified or overexpressed). HR-, HER2+ metastatic breast cancer is treated with HER2-targeted therapy and other systemic therapies that may include chemotherapy or antibody-drug conjugates. See **Guide 5**.

Guide 5
Systemic therapy options for HER2+

First-line therapies	• Docetaxel plus Pertuzumab plus Trastuzumab (preferred). If HR+, then aromatase inhibitor with or without Palbociclib plus Pertuzumab plus Trastuzumab • Paclitaxel plus Pertuzumab plus Trastuzumab (preferred). If HR+, then aromatase inhibitor with or without Palbociclib plus Pertuzumab plus Trastuzumab • Fam-trastuzumab deruxtecan-nxki (T-DXd) plus Pertuzumab (other recommended)
Second- and third-line therapies	• Capecitabine and Tucatinib plus Trastuzumab (preferred) • Fam-trastuzumab deruxtecan-nxki (T-DXd) (preferred) • Ado-trastuzumab emtansine (T-DM1)
Fourth-line therapies and beyond	• Docetaxel or Vinorelbine plus Trastuzumab • Paclitaxel plus Trastuzumab with or without Carboplatin • Capecitabine and Lapatinib or Capecitabine plus Trastuzumab • Lapatinib plus Trastuzumab • Other chemotherapies plus Trastuzumab • Capecitabine and Neratinib • Chemotherapy (Capecitabine, Eribulin, Gemcitabine, or Vinorelbine) plus Margetuximab-cmkb • Abemaciclib and Fulvestrant plus Trastuzumab (for HR+ only) • Targeted therapy might be used if specific mutations are found

HR- and HER2- metastatic breast cancer

HR-, HER2- metastatic breast cancer is also called triple-negative breast cancer (TNBC). In TNBC, cancer cells test negative for HER2, estrogen receptors, and progesterone receptors. It's written as ER- and/or PR- with HER2-.

There are many variations within TNBC. It's a group of diseases that we are learning more about all the time. Treatment is usually chemotherapy, and sometimes immunotherapy or targeted therapy.

TNBC systemic therapy options are listed in **Guide 6**.

Monitoring

You'll be monitored throughout treatment. Monitoring includes physical exams, blood tests, imaging scans, and tumor testing (if needed). Monitoring is used to see if your cancer is responding to treatment, stable, or progressing.

During this time, you'll also be monitored for symptoms caused by cancer, such as pain from bone metastases. The goal of monitoring is to determine if treatment provides benefit. Benefit includes keeping cancer stable.

Keep these things in mind:

- Tell your care team about any symptoms, such as headaches, menstrual spotting between periods or new onset of spotting after menopause (if you've taken tamoxifen in the past), shortness of breath when walking, or bone pain. Side effects can be managed.
- Continue to take all medicine, such as endocrine therapy, exactly as prescribed—don't miss or skip doses. If you're experiencing side effects, talk to your care team.

Keep follow-up visits and imaging test appointments. Seek good routine medical care, including regular doctor visits for preventive care and cancer screening.

Survivorship resources

More information on supportive care is available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

Guide 6
Systemic therapy options for TNBC

Preferred first-line therapies	• Single-agent chemotherapy, such as Doxorubicin, Liposomal Doxorubicin, Paclitaxel, Capecitabine, Gemcitabine, Vinorelbine, or Eribulin • For PD-L1 with a combined positive score (CPS) of 10 or more, chemotherapy (Albumin-bound Paclitaxel, Paclitaxel, or Gemcitabine with Carboplatin) plus Pembrolizumab or Sacituzumab govitecan-hziy plus Pembrolizumab • For PD-L1 with a CPS of less than 10, Sacituzumab govitecan-hziy, Datopotamab deruxtecan-dlnk, or systemic chemotherapy • For PD-L1 with a CPS of 10 or more and germline *BRCA1* or *BRCA2* mutation, PARP inhibitors (Olaparib or Talazoparib) or platinum chemotherapy (Cisplatin or Carboplatin)
Second-line therapies and beyond	• For germline *BRCA1* or *BRCA2* mutation, PARP inhibitors (PARPi) (preferred) • Sacituzumab govitecan-hziy (preferred) if not used before • Fam-trastuzumab deruxtecan-nxki (T-DXd) (other recommended) • Chemotherapy or targeted therapy
Other recommended therapies	• Albumin-bound Paclitaxel • Cyclophosphamide • Docetaxel • Epirubicin • Ixabepilone • For germline *PALB2,* Olaparib
Therapies used in certain cases	• Doxorubicin and Cyclophosphamide (AC) • Epirubicin and Cyclophosphamide (EC) • Cyclophosphamide, Methotrexate, and Fluorouracil (CMF) • Capecitabine and Docetaxel • Gemcitabine and Paclitaxel (GT) • Carboplatin and Gemcitabine • Carboplatin and Albumin-bound Paclitaxel or Paclitaxel • Targeted therapies for *NTRK* fusion, MSI-H/dMMR, TMB-H, *RET* fusion, somatic *BRCA1* or *BRCA2* mutations, *FGFR1-3* fusions or mutations, and HER2 activating mutations
Notes	• Alternative taxanes (such as docetaxel, paclitaxel, albumin-bound paclitaxel) might be substituted in some cases • Targeted therapy might be used if specific mutations are found

Living with metastatic breast cancer

A person is a cancer survivor from the time of diagnosis until the end of life. People living with metastatic breast cancer often have unique challenges not experienced by other breast cancer survivors. Living with metastatic breast cancer can be stressful—financially, emotionally, physically, and spiritually. Your cancer care team or one of your primary health care providers can be a great source of information and support if you're experiencing any changes in your mental, physical, emotional, sexual, or financial health.

Lifestyle changes

Changes in your lifestyle can help manage side effects and improve treatment results. Lifestyle changes include eating a balanced and nutritious diet, exercising regularly, limiting alcohol, and quitting smoking. Your care team can offer information and support on lifestyle changes. Optimizing your weight is an important aspect of managing your overall health and it can improve breast cancer outcomes.

Disease progression

Disease progression is defined by the growth or spread of cancer as shown on imaging tests or physical exam of the tumor. When a therapy no longer slows tumor growth, the current therapy is stopped and another one is started. We often call each therapy a line of therapy. Before each new line of therapy, you and your care team will discuss the risks and benefits of treatment, your overall health, and your goals for treatment.

Systemic therapy has side effects that may impact your quality of life. If you and your care team decide to stop systemic therapy, supportive care will continue. Supportive care is sometimes called palliative care.

Your preferences and goals of treatment are always important. The goals of treatment can be thought of as curative or palliative. This means that treatment aims to cure or, in the case of palliative care, to preserve function and/or maintain comfort. Goals of treatment may change over time. Talk with your care team and make your wishes known.

Progression or toxicity on endocrine therapy

First-line therapy is the first treatment given. If HR+ cancer progresses while you're on first-line endocrine therapy, then you'll likely switch to a different endocrine therapy. Targeted therapy might be added. Sometimes, cancer becomes resistant to endocrine therapy. If this happens, endocrine therapy will be stopped and another systemic therapy, such as chemotherapy or an antibody-drug conjugate, will be used.

Progression on systemic therapy with HER2-targeted therapy

If HER2+ cancer progresses while you're on HER2-targeted therapy, then different HER2-targeted therapy will be given.

Key points

- In hormone receptor-positive (HR+) cancer, estrogen receptors (ER+), progesterone receptors (PR+), or both (ER+/PR+) are found.
- In HER2-positive (HER2+) cancer, HER2 is either overexpressed or amplified. HER2+ cancer is treated with HER2-targeted therapy and other systemic therapies.
- Initial treatment for HR+ with HER2- is most often endocrine therapy plus a CDK4/6 inhibitor.
- People with HR+ and HER2+ disease may also benefit from treatment with both endocrine therapy and HER2-targeted therapies.
- In triple-negative breast cancer (TNBC), receptors for estrogen, progesterone, and HER2 aren't found. TNBC is usually treated with chemotherapy and sometimes immunotherapy or targeted therapy.
- Before each new line of therapy, you and your care team will discuss the risks and benefits of treatment, your overall health, and your goals for treatment.
- It's important to keep follow-up visits and imaging test appointments. Seek good routine medical care, including preventive care and cancer screenings. Continue to take all medicines as prescribed.

Questions to ask

- Which systemic therapy do you recommend and why?
- What role do previous treatments, my age, menopause status, and overall health play in my treatment options?
- Does the order of treatments matter?
- How long will I be on this systemic therapy or endocrine therapy?
- How often will I have blood and imaging tests?

What's next?

The next chapter provides information about resources and where you can get support.

6

Other resources

Want to learn more? Here's how you can get additional help.

What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- Finding a medical oncologist who's an expert in breast cancer
- Making treatment decisions
- The details of treatment and possible side effects
- Getting financial help
- Living with metastatic breast cancer

What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

Bag It
bagitcancer.org

Bone Marrow & Cancer Foundation
bonemarrow.org

Breast Cancer Alliance
breastcanceralliance.org

Breastcancer.org
breastcancer.org

Brem Foundation to Defeat Breast Cancer
bremfoundation.org

CanCare, Inc.
cancare.org

Cancer*Care*
cancercare.org

Cancer Hope Network
cancerhopenetwork.org

Cancer Survivor Care
cancersurvivorcare.org

DiepC Foundation
diepcfoundation.org

FORCE: Facing Our Risk of Cancer Empowered
facingourrisk.org

GPAC Global Patient Advocacy Coalition
gpacunited.org

HIS Breast Cancer Awareness
hisbreastcancer.org

Imerman Angels
imermanangels.org

Inflammatory Breast Cancer Research Foundation
ibcresearch.org

Lobular Breast Cancer Alliance
lobularbreastcancer.org

My Faulty Gene
myfaultygene.org

PAN Foundation
panfoundation.org

Sharsheret
sharsheret.org

Stupid Cancer
stupidcancer.org

Triage Cancer
triagecancer.org

Unite for HER
uniteforher.org

Young Survival Coalition (YSC)
youngsurvival.org

Questions to ask

- Who can I talk to about help with housing, food, and other basic needs?
- What help is available for transportation, childcare, and home care?
- What other services are available to me and my caregivers?
- How can I connect with others and build a support system?
- Who can I talk to if I don't feel safe at home, at work, or in my neighborhood?

"Be your own advocate. Talk to someone who has gone through the same thing as you. Ask a lot of questions, even the ones you are afraid to ask. You have to protect yourself and ensure you make the best decisions for you, and get the best care for your particular situation."

Words to know

antibody-drug conjugate (ADC)
A substance made up of a protein linked to a drug that attaches and enters certain types of cancer cells.

anti-estrogen
A drug that stops estrogen from attaching to cells.

aromatase inhibitor (AI)
A drug that lowers the level of estrogen in the body.

bilateral oophorectomy
An operation that removes both ovaries.

biopsy
A procedure that removes fluid or tissue samples to be tested for a disease.

bone mineral density
A test that measures bone strength.

bone scan
A test that makes pictures of bones to look for health problems

clinical trial
A type of research that assesses health tests or treatments.

contrast
A substance put into the body to make clearer pictures during imaging tests.

core needle biopsy
A procedure that removes tissue samples with a hollow needle. Also called core biopsy.

endocrine therapy
Cancer treatment that stops the making or action of estrogen. Also called hormone therapy.

estrogen
A hormone that plays a role in breast development.

estrogen receptor (ER)
A protein inside cells that binds to estrogen.

estrogen receptor-negative (ER-)
A type of breast cancer that doesn't use estrogen to grow.

estrogen receptor-positive (ER+)
A type of breast cancer that uses estrogen to grow.

gene
Coded instructions in cells for making new cells and controlling how they behave.

genetic counseling
Expert guidance on the chance for a disease that is passed down in families.

hereditary breast cancer
Breast cancer likely caused by abnormal genes passed down from biological parent to child.

histology
The structure of cells, tissue, and organs as viewed under a microscope.

hormone
A chemical in the body that triggers a response from cells or organs.

hormone receptor-negative cancer (HR-)
Cancer cells that don't use hormones to grow.

hormone receptor-positive cancer (HR+)
Cancer cells that use hormones to grow.

human epidermal growth factor receptor 2 (HER2)
A protein on the surface of a cell that sends signals for the cell to grow.

immunohistochemistry (IHC)
A lab test of cancer cells to find specific cell traits involved in abnormal cell growth.

in situ hybridization (ISH)
A lab test of the number of copies of a gene.

luteinizing hormone-releasing hormone (LHRH)
A hormone in the brain that helps control the making of estrogen by the ovaries.

lymph
A clear fluid containing white blood cells.

lymphedema
Swelling in the body caused by a buildup of fluid called lymph.

lymph node
A small, bean-shaped disease-fighting structure.

magnetic resonance imaging (MRI)
A test that uses radio waves and powerful magnets to take pictures of the inside of the body.

medical oncologist
A doctor who is an expert in cancer drugs.

menopause
12 months after the last menstrual period.

mutation
An abnormal change.

pathologist
A doctor who's an expert in testing cells and tissue to find disease.

perimenopause
The time during which your body makes the natural transition to menopause.

positron emission tomography (PET)
A test that uses radioactive material to see the shape and function of body parts.

postmenopause
The state of having no more menstrual periods.

premenopause
The state of having menstrual periods.

progesterone
A hormone involved in sexual development, periods, and pregnancy.

progesterone receptor (PR)
A protein inside cells that binds to progesterone.

selective estrogen receptor degrader (SERD)
A drug that blocks and destroys estrogen receptors.

selective estrogen receptor modulator (SERM)
A drug that blocks the effect of estrogen inside of cells.

side effect
An unhealthy or unpleasant physical or emotional response to treatment.

standard of care
The best-known way to treat a particular disease based on past clinical trials. There may be more than one treatment regimen that is considered standard of care.

supportive care
Health care that includes symptom relief but not cancer treatment. Also called palliative care or best supportive care.

systemic therapy
Drug treatment that works throughout the body.

triple-negative breast cancer (TNBC)
A breast cancer that does not use hormones or the HER2 protein to grow.

ultrasound
A test that uses sound waves to take pictures of the inside of the body.

visceral crisis
When organs aren't working as well as they should.

Let us know what you think!

Please take a moment to complete an online survey about the NCCN Guidelines for Patients.
NCCN.org/patients/response

NCCN Contributors

This patient guide is based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Breast Cancer, Version 3.2026. It was adapted, reviewed, and published with help from the following people:

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Boston, Massachusetts

Duke Cancer Institute
Durham, North Carolina

Fox Chase Cancer Center
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Fred & Pamela Buffett Cancer Center
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Rochester, Minnesota

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St. Jude Children's Research Hospital/
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Stanford Cancer Institute
Stanford, California

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Columbus, Ohio

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Sacramento, California

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Ann Arbor, Michigan

University of Wisconsin Carbone Cancer Center
Madison, Wisconsin

UT Southwestern Simmons
Comprehensive Cancer Center
Dallas, Texas

Vanderbilt-Ingram Cancer Center
Nashville, Tennessee

Yale Cancer Center/Smilow Cancer Hospital
New Haven, Connecticut

Notes

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NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

Metastatic Breast Cancer

To support the NCCN Guidelines for Patients, visit

NCCNFoundation.org/Donate

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